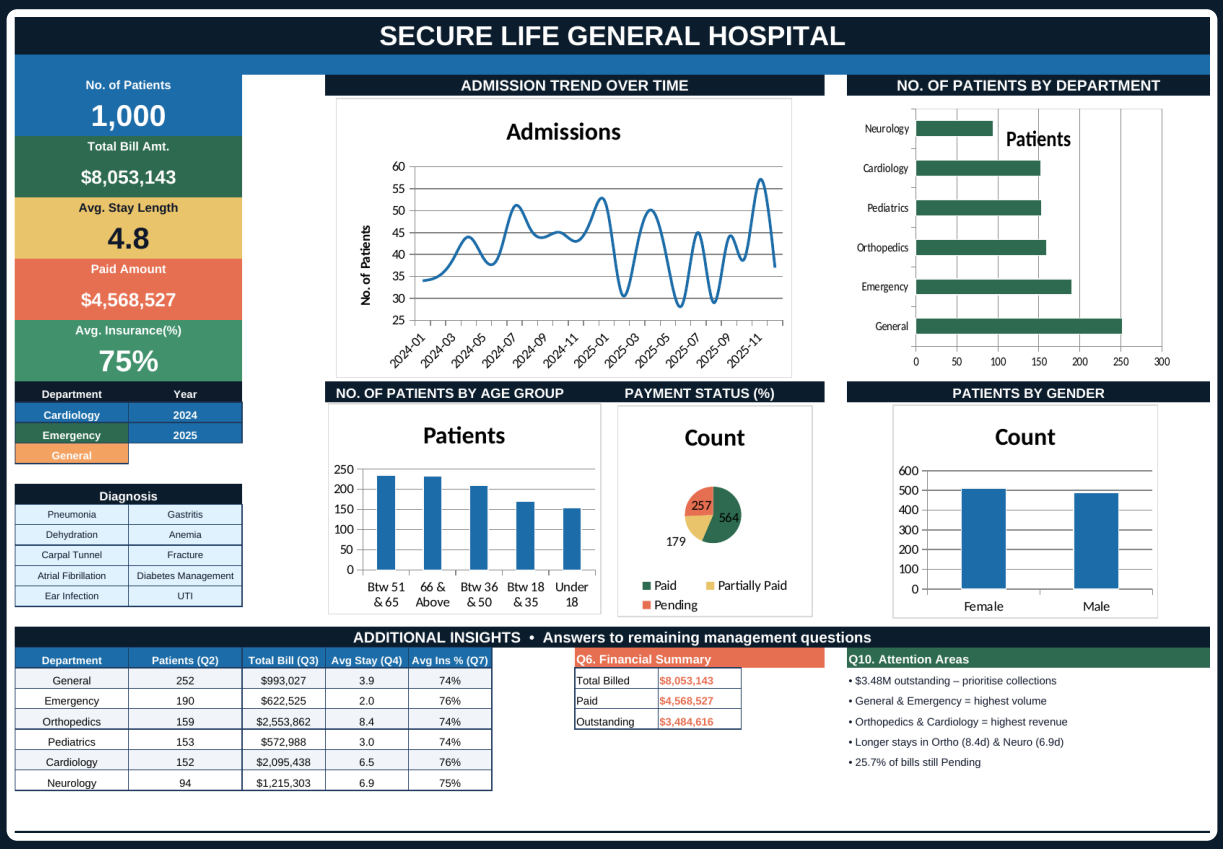
DATA ANALYTICS PROJECT · FSP COHORT 2

Build the Secure Life General Hospital Performance Dashboard

A complete step-by-step guide for absolute beginners in Microsoft Excel



32 numbered steps

9 charts

0 Excel experience needed

Prepared for Eunice Maafia Marfo · Patient Insights • Billing Analysis • Operational Excellence

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How to use this guide

You do **not** need any Excel experience. This guide takes you from an empty workbook to the finished **Secure Life General Hospital Performance Dashboard**, one small step at a time. Every number, colour, size and formula in it was checked against your finished workbook, so if you follow the steps in order you will end up with the same dashboard and the same answers.

Work in order. The Dashboard (Part 4) reads from the Summary sheet (Part 3), which reads from the Patient Data sheet (Part 2). If you skip ahead, the charts will have nothing to show.

The boxes you will see

> **TIP**
> A shortcut or a helpful habit.

> **WATCH OUT**
> A common mistake. Read these before you click.

> **CHECKPOINT**
> The exact result you should see. If yours differs, stop and fix it before moving on.

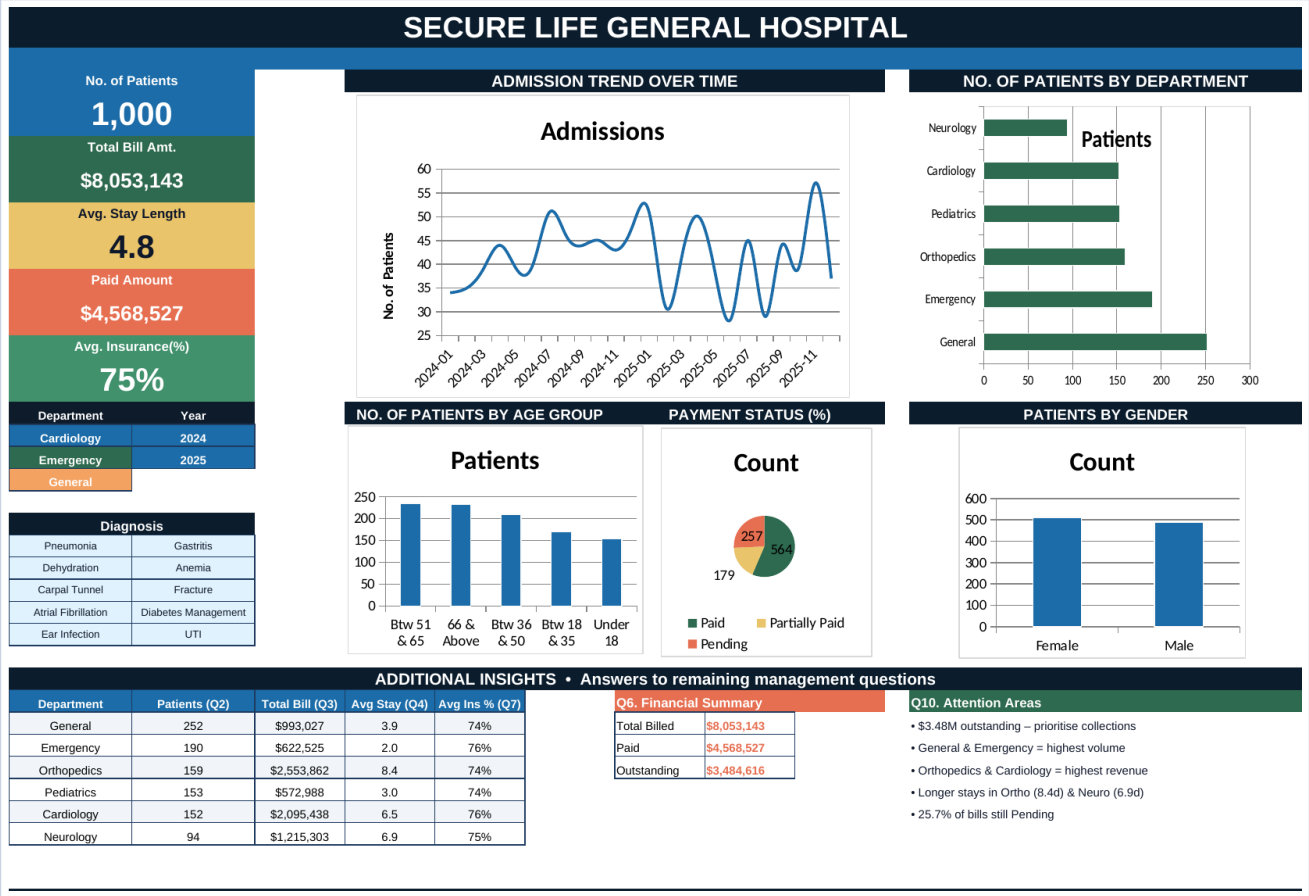
> **WHY THIS WORKS**
> A plain-English explanation, so you learn instead of just copying.

How instructions are written

You will see	It means
Home ▸ Merge & Center	Click the **Home** tab on the ribbon, then the **Merge & Center** button.
`Ctrl` + `1`	Hold the first key and press the second. **Mac users:** use `Cmd` instead of `Ctrl`.
`=SUM(A1:A5)`	Something you **type** into a cell exactly as shown (blue box).
`#1B6CA8`	A colour code. In the colour picker choose **More Colors > Custom** and paste it in the **Hex** box. (No Hex box? Use the RGB list in Appendix A.)
B3:C5	A **range**: every cell from B3 down to C5. Click B3, hold `Shift` and click C5 to select it.

> **GOOD TO KNOW**
> **What you need:** desktop Microsoft Excel (Microsoft 365, 2019 or 2021, on Windows or Mac), your hospital dataset (1,000 visit records, columns A to K), and about **4 to 5 hours** at a relaxed beginner pace. Menu names below follow Windows Excel; Mac Excel is nearly identical.
>
> You can split the work into four sessions: **Session 1** Parts 1–2 (data), **Session 2** Part 3 (summary), **Session 3** Part 4, Steps 13–25 (dashboard), **Session 4** Steps 26–32 (Deeper Dive row). **Save with** `Ctrl` + `S` **after every step.**
>
> **Short on time?** Open **Practice_Kit/Practice_Workbook_START_HERE.xlsx**: Steps 1 and 2 are already done for you.

What you are going to build



The finished dashboard (rendered from your workbook). Every part of this picture is built in this guide.

The dashboard answers **ten questions** that hospital management asked. Here is where each answer lives, and which step builds it:

#	Management question	Where the dashboard answers it	Step
1	How many patient visits?	KPI card “No. of Patients” (1,000)	15
2	Which departments get the most patients?	Bar chart “No. of Patients by Department” + Patients (Q2) column	19, 23
3	Which departments bill the most?	Total Bill (Q3) column + column chart “Total Bill by Department”	23, 27
4	Average length of stay by department?	Avg Stay (Q4) column + column chart “Avg Stay by Department”	23, 28
5	Share of bills Paid / Partially Paid / Pending?	Pie chart “Payment Status”	21
6	How much billed, how much outstanding?	KPI cards + “Q6. Financial Summary” box	15, 24
7	How does insurance coverage vary?	Avg Ins % (Q7) column + KPI card + chart “Avg Insurance Coverage by Department”	15, 23, 29
8	Which age groups visit most?	Column chart “No. of Patients by Age Group”	20
9	How do admissions change over time?	Line charts “Admission Trend Over Time” and “Billing Trend”	18, 30
10	What needs management attention?	“Q10. Attention Areas” box	25

GOOD TO KNOW

What the finished workbook contains. Five KPI cards (the Outstanding amount sits in the Q6 box) and **nine charts**: the five main charts shown above (Steps 18 to 22) plus a “Deeper Dive” row of four more underneath (Steps 27 to 30). This guide builds **all of them**, exactly as described in your project write-up.

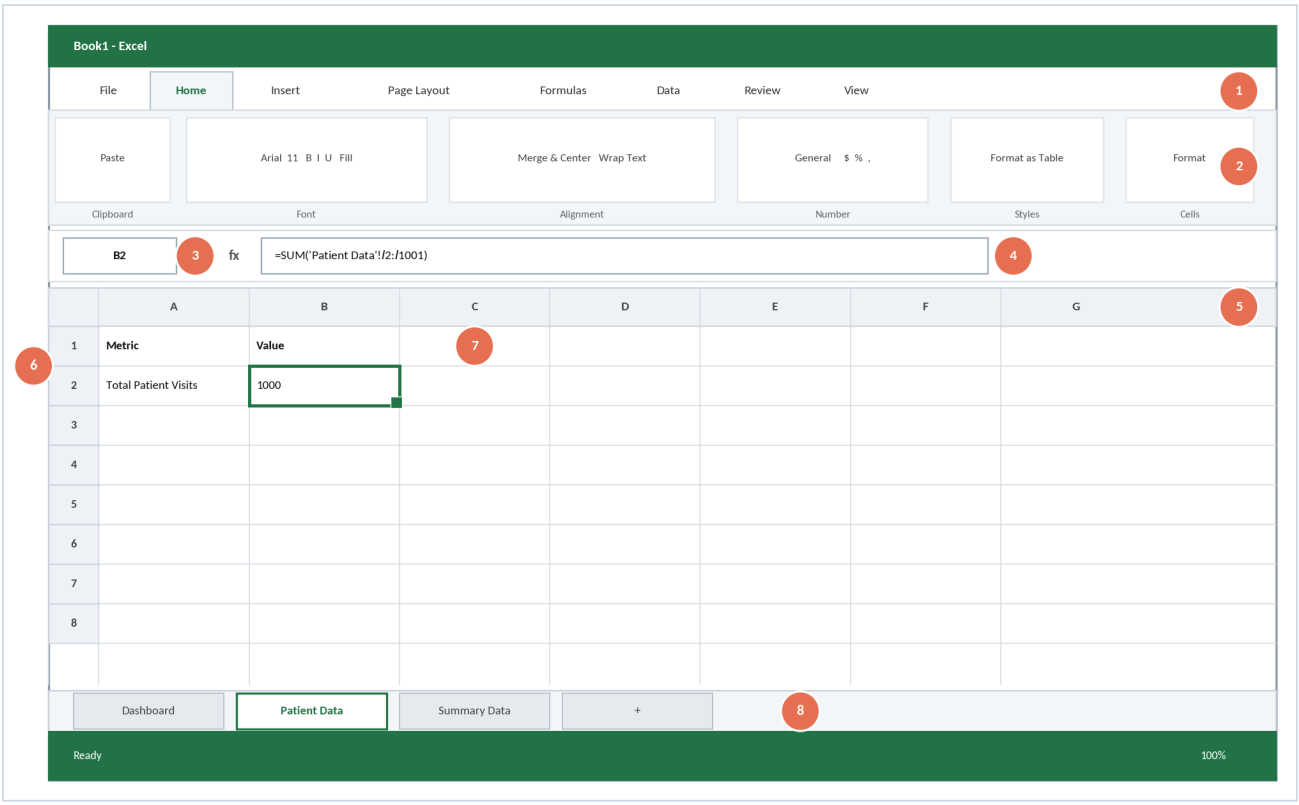
PART 1

Get Ready

A 10-minute tour of Excel, the key ideas, and the blueprint of the dashboard.

about 25 min reading

Meet the Excel window



A simplified Excel window. Your screen will look slightly different, but these eight parts are always there.

	Part	What it does
1	Ribbon tabs	File, Home, Insert, Page Layout, Formulas, Data, Review, View. Formatting lives mostly on **Home**; charts live on **Insert**.
2	Ribbon	The buttons for the tab you picked, arranged in groups (Font, Alignment, Number…).
3	Name Box	Shows the **address** of the selected cell (here B2). Type an address, such as B30, and press Enter to jump there.
4	Formula bar	Shows what is really inside the selected cell (a formula, or just text). Click here to edit.
5	Column letters	A, B, C… Right-click a letter to change that column's width.
6	Row numbers	1, 2, 3… Right-click a number to change that row's height.
7	Active cell	The green box = the cell you are working in. The tiny square in its corner is the **fill handle** (used to copy formulas down).
8	Sheet tabs	One workbook holds many sheets. You will use three: **Dashboard**, **Patient Data**, **Summary Data**.

Ten ideas that make everything else easy

Idea	What it means	Example
1. Cell address	Column letter + row number.	**B2** = column B, row 2
2. Range	A block of cells written *first*:*last*.	**A2:A1001** = 1,000 cells down column A
3. Formula	Anything that starts with **=**. Excel calculates it and updates it automatically if the data changes.	`=F2-E2`
4. Function	A ready-made calculator with a name and brackets.	`=SUM(I2:I1001)`
5. Relative reference	A reference that shifts when you copy the formula down a row.	Copy `=F2-E2` down one row and it becomes `=F3-E3`
6. Absolute reference	Dollar signs **$** freeze a reference so it does *not* shift. Press `F4` while editing to add them. ($A10 freezes only the column.)	`$I$2:$I$1001`
7. Other sheets	Put the sheet name in quotes, then **!**, then the cell. Quotes are needed when the name has a space.	`'Patient Data'!I2`
8. Fill handle	Drag the small square at a cell's corner, or **double-click** it, to copy the cell downwards.	Double-click = copies to the last row of data
9. Value vs display	What a cell *shows* can differ from what it *holds*. Number formats change the look, not the number.	75 can be shown as 75% with the format `0"%"`
10. Merge cells	Joins several cells into one big cell. Great for banners; use it **only on the Dashboard**, never on data.	Home ▸ Merge & Center

Typing a formula safely

1. Click the cell, then type **=**.
2. Start typing a function name (for example **SUM**). A list appears; press `Tab` to accept the suggestion.
3. To point at cells instead of typing their addresses, just **click** them (even on another sheet) – Excel writes the address for you.
4. Close any open bracket, then press `Enter`. Press `F2` to edit the cell again, or `Esc` to cancel.

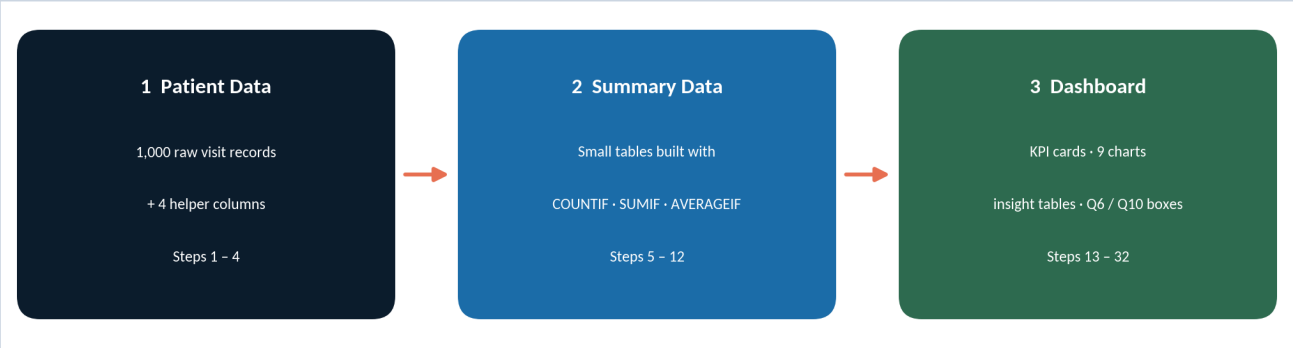
> **WATCH OUT**
>
> Text inside formulas must be wrapped in **straight double quotes**, like `"Paid"`. Use the keyboard quote key, not curly quotes copied from a Word document, or Excel shows **#NAME?**.
>
> In some countries Excel uses **semicolons** instead of commas between function inputs (for example `=IF(A1>1;"a";"b")`). If a comma formula is rejected, try semicolons.

Your five golden rules

1	**Save often** (`Ctrl` + `S`). Excel is not like a phone app that saves for you.
2	**Undo is your friend** (`Ctrl` + `Z`). Nothing you try here can break your original data file.
3	**Check every CHECKPOINT.** A wrong number found early is a 10-second fix; found late, it is a 2-hour hunt.
4	**Spell labels exactly** (capital letters, spaces, &). "Partially Paid" and "Partially paid" can behave differently in charts and tables.
5	**Keep raw data raw.** Never type over the original 11 columns; all calculations go in new columns or other sheets.

The blueprint: how the workbook is organised

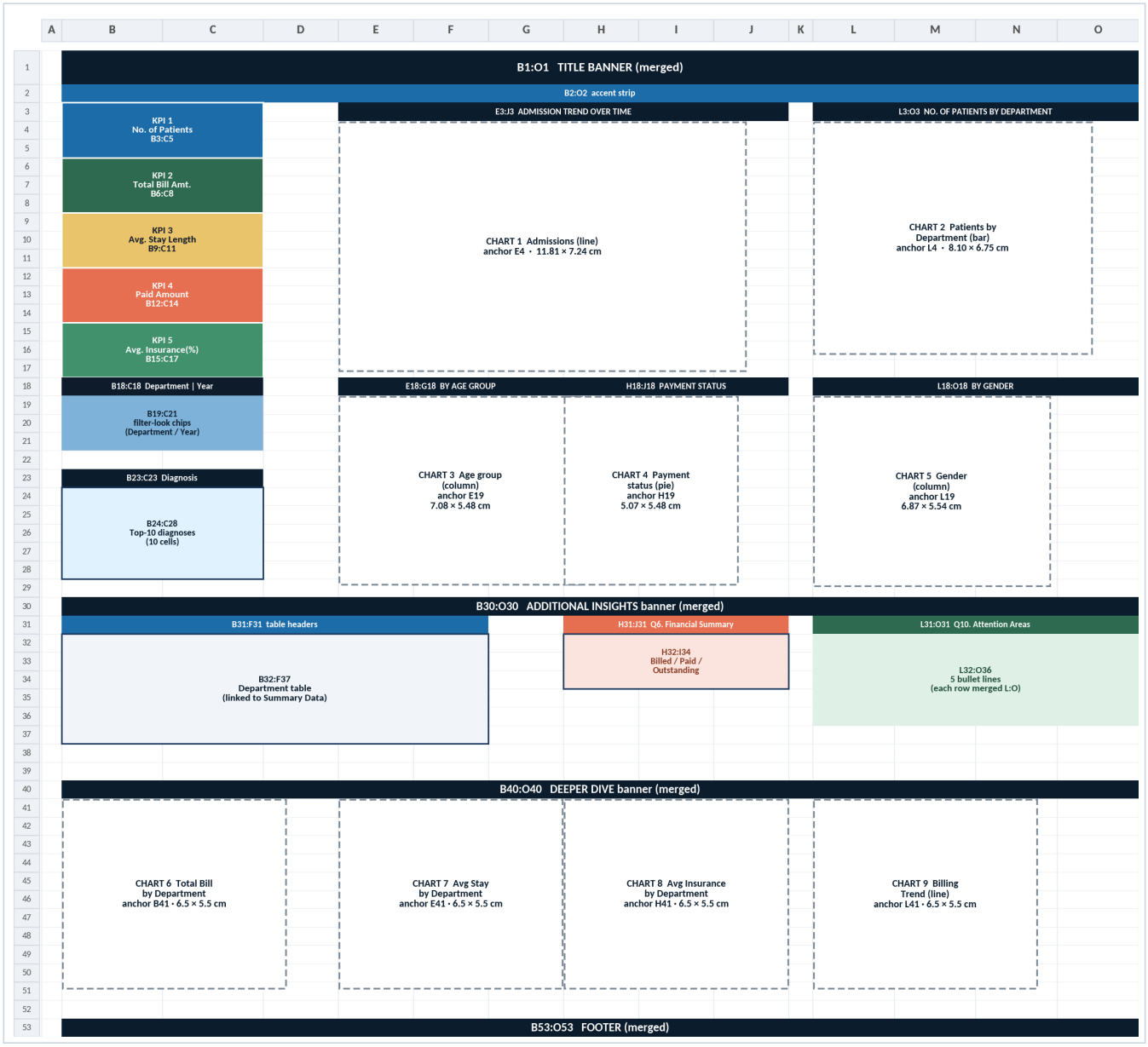
Professional dashboards are built in **layers**. The raw data is kept on one sheet, small calculation tables on a second sheet, and the pretty presentation on a third. Charts and cards on the Dashboard simply *read* the Summary tables, so when data changes, everything updates.



The three layers of your workbook (sheet order left to right in the file: Dashboard, Patient Data, Summary Data).

The Dashboard sheet map

Every box below is built in the order of the steps. The small grey letters and numbers are the real Excel column letters and row numbers, so you can build cell-for-cell. Dashed boxes are charts, with the cell each chart's top-left corner sits on and its size.



Cell map of the Dashboard sheet (columns A–O, rows 1–38). Keep this page handy while building Part 4.

PART 2

The Data Sheet

Create the workbook, bring in the 1,000 patient records, check them, and add four helper columns.

Steps 1 – 4 • about 45 min

STEP 1 Create the workbook and its three sheets

5 min

1. Open Excel and choose **Blank workbook**.
2. Save it straight away: **File ▸ Save As**, choose a folder, name it **SECURE_LIFE_GENERAL_HOSPITAL_Dashboard** and keep the type **Excel Workbook (*.xlsx)**.
3. **Double-click** the tab called *Sheet1*, type **Dashboard** and press `Enter`.
4. Click the **+** button next to the tabs to add a sheet. Double-click its tab and name it **Patient Data**.
5. Add one more sheet and name it **Summary Data**. Drag the tabs (if needed) so the order is: **Dashboard, Patient Data, Summary Data**.

> **WATCH OUT**
> Sheet names must be spelled exactly like this (capital letters and the space). Every formula in this guide refers to them by name.

STEP 2 Bring in the 1,000 patient records

10 min

Your dataset has **1,000 rows** (one per patient visit) and **11 columns**, A to K. Here is what each field contains:

Col	Field	Type	Example (row 2)	Used for
A	Patient ID	Text	PAT-3001	Unique visit ID (counting visits)
B	Patient Name	Text	Thomas Martin	Identification only
C	Age	Number	57	Age groups
D	Gender	Text	Male	Gender chart
E	Admission Date	Date	2024-07-15	Stay length, month, year
F	Discharge Date	Date	2024-07-23	Stay length
G	Department	Text	Cardiology	Department table / chart
H	Diagnosis	Text	Atrial Fibrillation	Top-10 diagnoses
I	Total Bill Amount	Number	15359.41	Billing totals
J	Insurance Coverage %	Number	60	Average insurance
K	Payment Status	Text	Paid	Paid / Pending / Partially Paid

1. Open the file that holds your 1,000 records. Click any data cell and press `Ctrl` + `A` to select everything, then `Ctrl` + `C` to copy.
2. Go back to your new workbook, open the **Patient Data** tab, click cell **A1** and press `Ctrl` + `V`. (Starting from a CSV file instead? Use **Data ▸ From Text/CSV** then **Load**.)
3. In **L1, M1, N1, O1** type the four new column names exactly: **Stay_Length**, **Year**, **YearMonth**, **Age_Group**. (Note the underscore in Stay_Length and Age_Group.) You will fill them in Step 4.

4. **Style the header row.** Select **A1:O1**. Fill colour `#0D1B2A` (**Home ▸ Fill Color**), font **Arial 10, bold, white**.
5. **Widen the columns.** Click the grey corner above row 1 to select all columns, then **Home ▸ Format ▸ Column Width** and type **14**.
6. **Format the date columns.** Select columns **E:F**, press `Ctrl` + `1`, choose **Number > Custom** and type `yyyy-mm-dd`.

CHECKPOINT

Cell **A2** = PAT-3001, **B2** = Thomas Martin, **K2** = Paid. The **last row (1001)** must be **PAT-4000, Jennifer Ramirez**.

Dates in E and F are **right-aligned**. Left-aligned dates are really text and will break Step 4 (see Appendix D).

TIP

Want the header to stay visible while you scroll 1,000 rows? **View ▸ Freeze Panes ▸ Freeze Top Row**.

STEP 3 Check the data quality

10 min

Analysts always inspect data before trusting it. Do these four quick checks; each should come out clean.

Check	How	Expected result
Row count	In an empty cell (say Q1) type `=COUNTA(A2:A1001)` then delete it afterwards.	**1000**
Duplicate IDs	Click any cell in column A > **Data ▸ Remove Duplicates**, tick **only** column A.	"No duplicate values found" (1,000 unique IDs)
Blank cells	Select **A1:K1001** > **Home ▸ Find & Select ▸ Go To Special** > **Blanks**.	"No cells were found"
Spelling of labels	Select A1:K1001 > **Data ▸ Filter**. Open the arrows on **Department**, **Gender** and **Payment Status**, then switch the filter off again.	Department: Cardiology, Emergency, General, Neurology, Orthopedics, Pediatrics. Gender: Female, Male. Payment Status: Paid, Partially Paid, Pending.

WHY THIS WORKS

A hidden typo such as "Paid " (with a trailing space) would be counted as a *different* status and quietly make every total wrong. Thirty seconds of checking now protects the whole dashboard.

STEP 4 Add the four helper columns (L to O)

20 min

The raw data does not contain *how long a patient stayed* or *which age band they belong to*. We calculate these once per row in four **helper columns**, so the summary tables can simply count them.

4a Stay_Length – column L

1. Click **L2** and type the formula below, then press `Enter`.

L2 (days between admission and discharge)

```
=F2-E2
```

Subtracting two dates gives the number of days between them. Row 2 should give **8** (15 July to 23 July).

WATCH OUT

Excel sometimes displays the answer as a strange **date or time** (for example 1900-01-08). The number is right; only the *format* is wrong. Click L2 and set **Home ▸ Number Format box ▸ General** (or **Number** with 0 decimals), then copy the format down with the next step.

2. Copy the formula to all rows: click **L2**, move the mouse over the small square at its bottom-right corner (the fill handle) until the pointer becomes a thin **+**, then **double-click**. Excel fills down to row 1001.

4b Year – column M

M2 (pulls the year out of the admission date)

```
=YEAR(E2)
```

4c YearMonth – column N

N2 (turns the date into text like 2024-07)

```
=TEXT(E2,"yyyy-mm")
```

WHY THIS WORKS

TEXT(date, format) converts a date to text in the layout you choose. Writing the month as **yyyy-mm** means the labels sort in true time order (2024-01, 2024-02, …) and make tidy chart labels.

If your Excel uses another language, the letters inside the quotes may differ (for example JJJJ-MM in German).

4d Age_Group – column O

O2 (type it as ONE line; it is wrapped here only to fit the page)

```
=IF(C2<18,"Under 18",IF(C2<=35,"Btw 18 & 35",
   IF(C2<=50,"Btw 36 & 50",IF(C2<=65,"Btw 51 & 65","66 & Above"))))
```

If Age is…	Excel writes…
under 18	Under 18
18 to 35	Btw 18 & 35
36 to 50	Btw 36 & 50
51 to 65	Btw 51 & 65
66 or more	66 & Above

WHY THIS WORKS

IF(test, result-if-true, result-if-false) asks a yes/no question. Our formula asks several in order: “Is the age under 18? If not, is it 35 or less? …” The first question answered **yes** wins, so each band only needs its upper limit.

Fill the last three columns down: select **M2:O2** together and double-click the fill handle to fill rows 2 to 1001 (check that column L was filled too).

CHECKPOINT

Compare these four rows with your sheet:

Row 2 → **8 | 2024 | 2024-07 | Btw 51 & 65**

Row 3 → **10 | 2025 | 2025-03 | Btw 51 & 65**

Row 4 → **3 | 2024 | 2024-04 | 66 & Above**

Row 1001 → **4 | 2024 | 2024-02 | Btw 18 & 35**

Quick sanity test: the smallest Stay_Length is **1** and the largest is **14**.

GOOD TO KNOW

Your original workbook stores these four columns as plain values. Keeping them as **formulas** is fine (they give identical results and recalculate if data changes). To freeze them as values: select L2:O1001, copy, then **Home ▸ Paste ▸ Paste Values**.

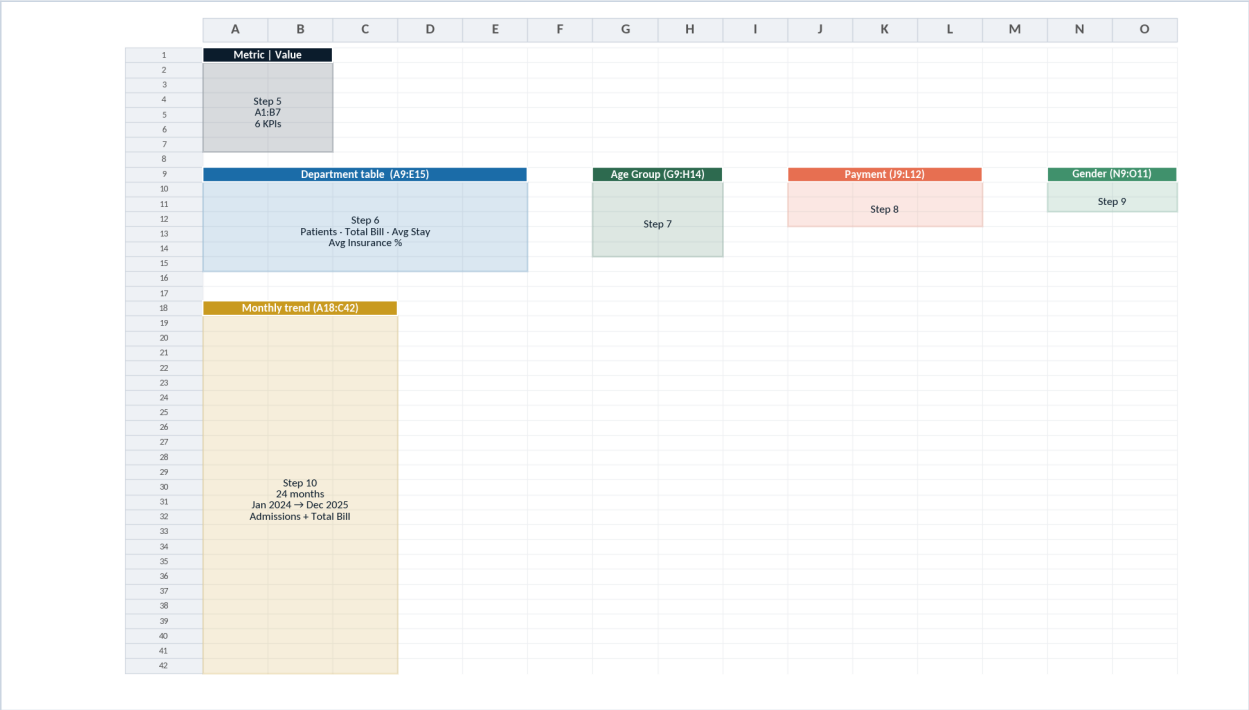
PART 3

The Summary Sheet

Build six small calculation tables with COUNTIF, SUMIF and AVERAGEIF. The dashboard will read these.

Steps 5 – 12 · about 60 min

Open the **Summary Data** tab. This sheet is the dashboard's engine room. It holds six little tables, each answering one question. Here is the layout you are about to create:



Where each table goes on the Summary Data sheet.

Ranges you will use again and again

Each formula points at a column of the Patient Data sheet, rows 2 to 1001. Fixing the range with **$** signs (absolute references) lets you copy the formula down without the range sliding.

Column	Meaning	Range to use
A	Patient ID	`'Patient Data'!$A$2:$A$1001`
C	Age	`'Patient Data'!$C$2:$C$1001`
D	Gender	`'Patient Data'!$D$2:$D$1001`
G	Department	`'Patient Data'!$G$2:$G$1001`
I	Total Bill Amount	`'Patient Data'!$I$2:$I$1001`
J	Insurance Coverage %	`'Patient Data'!$J$2:$J$1001`
K	Payment Status	`'Patient Data'!$K$2:$K$1001`
L	Stay_Length	`'Patient Data'!$L$2:$L$1001`
N	YearMonth	`'Patient Data'!$N$2:$N$1001`
O	Age_Group	`'Patient Data'!$O$2:$O$1001`

TIP

Typing those ranges by hand is slow. Instead, type `=SUM(`, click the **Patient Data** tab, drag down the column from row 2 to row 1001, press `F4` to add the $ signs, then finish the formula. Excel writes the range for you.

STEP 5 The six headline numbers (A1:B7)

8 min

Type the labels, then the formulas. **Start at cell A1.**

Cell	Type this	What it does	Result
A1 / B1	`Metric / Value`	Column headings	–
A2 / B2	`Total Patient Visits` `=COUNTA('Patient Data'!$A$2:$A$1001)`	Counts the filled Patient IDs	1000
A3 / B3	`Total Bill Amount` `=SUM('Patient Data'!$I$2:$I$1001)`	Adds every bill	8,053,143.44
A4 / B4	`Paid Amount` `=SUMIFS('Patient Data'!$I$2:$I$1001,'Patient Data'!$K$2:$K$1001,"Paid")`	Adds bills **only where** Payment Status is Paid	4,568,527.32
A5 / B5	`Outstanding Amount` `=B3-B4`	Total minus Paid	3,484,616.12
A6 / B6	`Avg Stay Length` `=ROUND(AVERAGE('Patient Data'!$L$2:$L$1001),1)`	Average days, rounded to 1 decimal	4.8
A7 / B7	`Avg Insurance %` `=ROUND(AVERAGE('Patient Data'!$J$2:$J$1001),0)`	Average coverage, whole number	75

WHY THIS WORKS

COUNTA counts non-empty cells. **SUM** adds. **SUMIFS(add-these, look-in-this, for-this-value)** adds only the rows that match a condition. **AVERAGE** finds the mean and **ROUND(number, decimals)** tidies it.

How "Outstanding" is defined here: everything not marked Paid, i.e. the full bills of *Partially Paid* and *Pending* visits ($1.40M + $2.08M). The data does not say how much of a partial bill was settled, so the dashboard treats the whole bill as still open. Mention this if someone asks.

STEP 6 Department table (A9:E15)

10 min

Headings in **A9:E9**: **Department, Patients, Total Bill, Avg Stay, Avg Insurance %**. Then type the six department names in **A10:A15**, in this order (largest to smallest by patients): **General, Emergency, Orthopedics, Pediatrics, Cardiology, Neurology**. Now type the four formulas in row 10:

Row 10 (do not type the cell names B10, C10…)

```
B10 =COUNTIF('Patient Data'!$G$2:$G$1001,$A10)
C10 =SUMIF('Patient Data'!$G$2:$G$1001,$A10,'Patient Data'!$I$2:$I$1001)
D10 =ROUND(AVERAGEIF('Patient Data'!$G$2:$G$1001,$A10,'Patient Data'!$L$2:$L$1001),1)
E10 =ROUND(AVERAGEIF('Patient Data'!$G$2:$G$1001,$A10,'Patient Data'!$J$2:$J$1001),1)
```

Select **B10:E10** and drag the fill handle down to row 15 (or double-click it). Because of the **$A10** in each formula, every row looks up its own department name.

Department	Patients	Total Bill	Avg Stay	Avg Insurance %
General	252	993,026.92	3.9	73.9
Emergency	190	622,525.28	2.0	76.2
Orthopedics	159	2,553,862.32	8.4	74.5
Pediatrics	153	572,987.96	3.0	74.0
Cardiology	152	2,095,437.97	6.5	76.3
Neurology	94	1,215,302.99	6.9	74.6

CHECKPOINT

Your table must match the one above. Patients add up to 1,000 (252+190+159+153+152+94).

Seeing **0** in a row? The name in column A does not exactly match the data (a stray space, or a different spelling). Re-type it.

WHY THIS WORKS

COUNTIF(where, what) counts matching rows. **SUMIF(where, what, add-these)** adds matching rows. **AVERAGEIF(where, what, average-these)** averages matching rows. The word *IF* in each name means "only the rows that match".

STEP 7 Age-group table (G9:H14)

4 min

Headings **G9: Age Group**, **H9: Patients**. Type these five labels down **G10:G14** exactly as written, then the formula in H10 and fill down to H14:

H10

```
H10 =COUNTIF('Patient Data'!$O$2:$O$1001,$G10)
```

G10:G14	Btw 51 & 65	66 & Above	Btw 36 & 50	Btw 18 & 35	Under 18
H10:H14	235	232	210	170	153

STEP 8 Payment-status table (J9:L12)

4 min

Headings **J9: Payment Status**, **K9: Count**, **L9: Amount**. Type **Paid, Partially Paid, Pending** in J10:J12, then fill down these two formulas:

Row 10

```
K10 =COUNTIF('Patient Data'!$K$2:$K$1001,$J10)
L10 =SUMIF('Patient Data'!$K$2:$K$1001,$J10,'Patient Data'!$I$2:$I$1001)
```

Payment Status	Count	Amount
Paid	564	4,568,527.32
Partially Paid	179	1,401,829.68
Pending	257	2,082,786.44

STEP 9 Gender table (N9:O11)

3 min

Headings **N9: Gender**, **O9: Count**. Type **Female** and **Male** in N10:N11, then:

O10 (fill down to O11)

```
O10 =COUNTIF('Patient Data'!$D$2:$D$1001,$N10)
```

CHECKPOINT

Female = **512**, Male = **488**. Age groups add to 1,000; Paid + Partially Paid + Pending = 564 + 179 + 257 = 1,000.

STEP 10

Monthly trend table (A18:C42)

12 min

This table feeds the Admissions line chart. Headings in **A18:C18**: **Year_Month, Admissions, Total Bill**. Then there are 24 rows, one per month from January 2024 to December 2025.

Month labels (A19:A42)

Type in A19, then fill down to A42

```
A19 =TEXT(EDATE(DATE(2024,1,1),ROW()-19),"yyyy-mm")
```

DATE(2024,1,1) is 1 January 2024. **EDATE(date, n)** moves a date forward by *n* months. **ROW()** is the current row number, so *ROW()-19* is 0 in row 19, 1 in row 20, and so on up to 23 in row 42. **TEXT** then writes each date as **yyyy-mm**.

TIP

Prefer typing the labels yourself? Type an apostrophe first: `'2024-01` . The apostrophe tells Excel "this is text". Without it Excel turns 2024-01 into a date such as **Jan-24**, which will not match your data.

Counts and bills per month (B19:C42)

Fill both formulas down to row 42

```
B19 =SUMPRODUCT(--('Patient Data'!$N$2:$N$1001=$A19))
C19 =SUMPRODUCT(('Patient Data'!$N$2:$N$1001=$A19)*'Patient Data'!$I$2:$I$1001)
```

WHY THIS WORKS

Why SUMPRODUCT and not COUNTIF? Labels like 2024-01 look like dates, and COUNTIF/SUMIF can misread them as dates. SUMPRODUCT compares the text *exactly*. Reading B19: the part in brackets is a list of TRUE/FALSE (does each row's YearMonth equal 2024-01?); the double minus **--** turns TRUE/FALSE into 1/0; SUMPRODUCT adds them up = number of admissions. In C19, the 1/0 list is multiplied by each bill before adding = total billed that month.

Answer key – all 24 months:

Month	Adm.	Total Bill	Month	Adm.	Total Bill
2024-01	34	271,784.38	2025-01	51	390,905.02
2024-02	35	309,160.45	2025-02	31	254,731.22
2024-03	39	263,013.64	2025-03	42	334,028.33
2024-04	44	314,674.54	2025-04	50	344,294.60
2024-05	39	366,399.52	2025-05	38	347,320.51
2024-06	40	389,933.45	2025-06	29	191,472.23
2024-07	51	421,474.20	2025-07	45	330,972.42
2024-08	46	361,232.66	2025-08	29	218,754.70
2024-09	44	299,681.50	2025-09	44	387,014.91
2024-10	45	443,607.11	2025-10	39	297,298.98
2024-11	43	317,696.95	2025-11	57	510,413.50
2024-12	48	373,806.96	2025-12	37	313,471.66

CHECKPOINT

Admissions in **B19:B42** add up to **1,000** and the bills in **C19:C42** add up to **8,053,143.44**. Peak month: **2025-11** with 57 admissions. Quietest: **2025-06** and **2025-08** with 29.

STEP 11 Find the top-10 diagnoses with a PivotTable

10 min

The Dashboard shows the ten most common diagnoses. A **PivotTable** is Excel's fastest way to count how often each value appears.

1. Open **Patient Data**, click any cell in the data, then **Insert ▸ PivotTable**. Confirm the range **A1:O1001** and choose **New Worksheet**, then **OK**.
2. In the field list on the right, drag **Diagnosis** into **Rows** and drag **Patient ID** into **Values** (it becomes "Count of Patient ID").
3. Right-click any number in the count column > **Sort** > **Sort Largest to Smallest**.
4. Read the top ten (below). You will type them into the Dashboard in Step 16. Afterwards you may delete this temporary sheet (right-click its tab > **Delete**).

Rank	Diagnosis	Visits
1	Pneumonia	43
2	Gastritis	39
3	Dehydration	38
4	Anemia	37
5	Carpal Tunnel	37
6	Fracture	37
7	Atrial Fibrillation	36
8	Diabetes Management	36
9	Ear Infection	34
10	UTI	33

> GOOD TO KNOW
> Ties (Carpal Tunnel and Fracture both 37; Atrial Fibrillation and Diabetes Management both 36) appear in alphabetical order, which is how the dashboard lists them.

STEP 12 Make the tables neat and run the sanity checks

10 min

Your original workbook formats each block as an Excel **Table** with a blue style. This is optional but makes the sheet easy to read.

1. Click inside the first block (A1:B7). Press `Ctrl` + `T`, tick **My table has headers**, click **OK**.
2. On the **Table Design** tab open **Table Styles** and pick **Blue, Table Style Medium 9** (hover to see names).
3. Repeat for **A9:E15**, **G9:H14**, **J9:L12**, **N9:O11** and **A18:C42**. Optionally untick **Filter Button** on the Table Design tab to hide the little arrows.
4. Widen columns so nothing shows ##### (Home > Format > AutoFit Column Width).

Sanity checks (type in a spare cell, read the answer, then delete)

Formula	Must equal
`=SUM(B10:B15)` `=SUM(H10:H14)` `=SUM(K10:K12)` `=SUM(O10:O11)` `=SUM(B19:B42)`	**1000** every time
`=SUM(C10:C15)` `=SUM(L10:L12)` `=SUM(C19:C42)`	**8053143.44** every time
`=B4+B5-B3`	**0**

CHECKPOINT

All sanity checks pass? Your Summary sheet is finished. **Save** (`Ctrl` + `S`). Everything on the Dashboard will now be linked to it.

PART 4

The Dashboard

Turn the numbers into a polished dashboard: KPI cards, panels, nine charts and insight boxes.

Steps 13 – 32 · about 130 min

Open the **Dashboard** tab. Keep the cell map from Part 1 nearby. You will build from the top-left to the bottom-right.

The colour palette

Navy	Blue	Dark green	Yellow	Coral	Green
#0B1C2C	#1B6CA8	#2D6A4F	#E9C46A	#E76F51	#40916C

More colours and RGB values are in Appendix A. All text on the dashboard uses the font **Arial**.

STEP 13 — Prepare the canvas

8 min

1. Click the **Dashboard** tab. Turn off the grid: **View** ▸ **Show** and **untick Gridlines**.
2. Set the column widths: select a column (or several by dragging across the letters), right-click a letter > **Column Width**, type the number, **OK**.
3. Set the row heights: click the grey corner (top-left) to select everything, **Home** ▸ **Format** ▸ **Row Height** = **15**. Then right-click row **1** > Row Height = **28**.

Columns	Width	Role
A	2.45	Left margin
B and C	15	KPI cards, panels, table
D	11	Spacer / Total Bill column
E to J	11	Chart zone 1
K	3	Spacer
L to O	12	Chart zone 2

TIP

Use the zoom slider at the bottom-right of Excel (around 80 to 100%) so you can read what you are building. A tiny zoom (for example 30%) makes the dashboard look empty when you reopen the file, so check it before saving.

STEP 14 — Title banner and accent strip

4 min

1. Select **B1:O1** and click **Home** ▸ **Merge & Center**. Type **SECURE LIFE GENERAL HOSPITAL** and press `Enter`.
2. With B1 selected: fill `#0B1C2C`, font **Arial**, size **20**, **bold**, white. Alignment: **Center** and **Middle Align**.
3. Select **B2:O2** (do not merge). Fill `#1B6CA8`. Leave it empty – it is a decorative strip.

STEP 15 — The five KPI cards (B3:C17)

20 min

A KPI (key performance indicator) card is a big number with a label. Each card is two merged blocks: a small **label** block on top and a large **value** block beneath, both two columns wide (B:C).

Card	Label cells (merge) & text	Value cells (merge) & link	Fill	Value font	Number format
1	B3:C3 No. of Patients	B4:C5 ='Summary Data'!B2	`#1B6CA8`	Arial 22 bold white	`#,##0` → 1,000
2	B6:C6 Total Bill Amt.	B7:C8 ='Summary Data'!B3	`#2D6A4F`	Arial 14 bold white	`$#,##0` → $8,053,143
3	B9:C9 Avg. Stay Length	B10:C11 ='Summary Data'!B6	`#E9C46A`	Arial 22 bold **dark** `#0F172A`	`0.0` → 4.8
4	B12:C12 Paid Amount	B13:C14 ='Summary Data'!B4	`#E76F51`	Arial 14 bold white	`$#,##0` → $4,568,527
5	B15:C15 Avg. Insurance(%)	B16:C17 ='Summary Data'!B7	`#40916C`	Arial 22 bold white	`0"%"` → 75%

How to build one card (repeat for all five)

1. Select the **label** range (for example B3:C3) > **Merge & Center**. Type the label text. Format: Arial **9**, bold, white (dark text for the yellow card), fill colour from the table, centred.
2. Select the **value** range (for example B4:C5) > **Merge & Center**. Click inside it and type `=`, then click the **Summary Data** tab and click the matching cell (for example B2), then press `Enter`. The Dashboard shows the value.
3. Format the value block: same fill colour, Arial size and colour from the table, centred horizontally and vertically.
4. Set the number format: select the value block, press `Ctrl` + `1`, choose **Number > Custom** and type the format code from the last column. **OK**.

WHY THIS WORKS

The formula **='Summary Data'!B2** simply says "show whatever is in Summary Data B2". If the data ever changes, the card updates by itself. Your original file typed these as plain text; linking is better practice and looks identical.

Custom number formats: #,##0 = thousands separator, no decimals; **$#,##0** adds a dollar sign; **0.0** = one decimal; **0"%"** shows a plain 75 with a % sign (without turning it into 7500%).

TIP

Finished card 1? Select its two merged blocks, click **Home** ▸ **Format Painter** and then click on the card 2 area to copy the layout, then only change the fill colour and the text. It saves a lot of clicks.

CHECKPOINT

Cards read, top to bottom: **1,000 · $8,053,143 · 4.8 · $4,568,527 · 75%.** (The sixth number from the report, Outstanding $3,484,616, appears later in the Q6 box.)

STEP 16 — Filter-look panels and the Diagnosis box

12 min

Below the cards are two small panels. The first (rows 18 to 21) is a **legend-style panel** showing the departments and years covered; the second (rows 23 to 28) lists the ten most common diagnoses.

Cell	Text	Fill	Font
B18, C18	Department, Year	`#0D1B2A`	Arial 8 bold white, centred
B19 / C19	Cardiology / 2024	`#1B6CA8`	Arial 8 bold white, centred
B20 / C20	Emergency / 2025	`#2D6A4F` / `#1B6CA8`	Arial 8 bold white, centred
B21	General	`#F4A261`	Arial 8 bold white, centred
B23:C23 (merge)	Diagnosis	`#0B1C2C`	Arial 9 bold white, centred
B24:C28 (10 cells)	The ten diagnoses (below)	`#E0F2FE`	Arial 7, dark `#0F172A`, centred

Type the diagnoses from Step 11 **left to right, row by row**:

	B	C
24	Pneumonia	Gastritis
25	Dehydration	Anemia
26	Carpal Tunnel	Fracture
27	Atrial Fibrillation	Diabetes Management
28	Ear Infection	UTI

1. Select **B19:C20** plus **B21** and **B24:C28** and add borders: **Home** ▸ **Borders** ▸ **All Borders**; set the line colour to `#1E3A5F` via **Borders** ▸ **Line Color** first, then apply.
2. For long names such as *Diabetes Management*: select B24:C28, press `Ctrl` + `1` > **Alignment** tab > tick **Shrink to fit**.

GOOD TO KNOW

The Department/Year chips are **fixed labels**, not working slicers; they make the dashboard look like a filter panel and show the period covered. Clicking them does nothing. (Part 5 mentions how to add real interactive slicers later.)

Type the **full** names ("Atrial Fibrillation", "Diabetes Management"); an earlier version of this dashboard cut them short.

STEP 17 — Section headings above each chart

6 min

Merge	Text	Fill / font
E3:J3	ADMISSION TREND OVER TIME	`#0B1C2C` · Arial 11 bold white
L3:O3	NO. OF PATIENTS BY DEPARTMENT	`#0B1C2C` · Arial 11 bold white
E18:G18	NO. OF PATIENTS BY AGE GROUP	`#0B1C2C` · Arial 10 bold white
H18:J18	PAYMENT STATUS (%)	`#0B1C2C` · Arial 10 bold white
L18:O18	PATIENTS BY GENDER	`#0B1C2C` · Arial 10 bold white

TIP

Do the first heading, then use **Format Painter** to copy it to the others, and retype the text.

Chart recipe (read once, use nine times)

1. **Select the data.** Go to the **Summary Data** sheet and select the block named in the step, **including its heading row**.
2. **Insert the chart.** Open the **Insert** tab and use the **Charts** group to pick the type named in the step. The chart appears on the Summary sheet.
3. **Move it to the Dashboard.** Click the chart's white border to select it, press `Ctrl` + `X`, open the **Dashboard** tab, click the **top-left cell** given in the step and press `Ctrl` + `V`. (It stays linked to the Summary data.)
4. **Size it.** Select the chart > **Format** tab (Chart Format / Shape Format) > **Size** group > type Height and Width. Sizes are given in cm and inches; Excel uses whichever your computer is set to.
5. **Place it.** Drag the chart by its border until its top-left corner sits on the anchor cell. Hold `Alt` while dragging to **snap** to the cell edges.
6. **Tidy with the + button.** Selecting a chart shows a green **+** (Chart Elements) beside it. Tick or untick **Chart Title**, **Axis Titles**, **Legend** and **Data Labels** there.
7. **Recolour.** Click once on a bar, line or slice to select the series, right-click > **Format Data Series** > **Fill & Line** (paint-bucket icon) > **Fill** > **Solid fill** > **Color** > **More Colors** > **Custom** and paste the Hex code.

WHY THIS WORKS

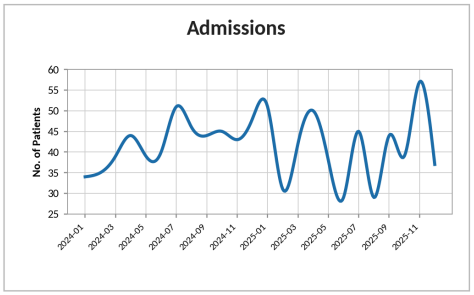
A chart title appears automatically from the heading cell of the data you selected (for example *Admissions*, *Patients*, *Count*). You can double-click a title to retype it. In your original workbook the titles were left as these automatic names.

STEP 18 Chart 1 – Admissions trend (line)

12 min

Select on Summary Data	Insert	Size (W × H)	Top-left at
A18:B42 (Year_Month, Admissions)	Charts > Insert Line or Area Chart > **2-D Line**	11.81 × 7.24 cm (4.65 × 2.85 in)	E4

1. Select **A18:B42** and insert a **2-D Line** chart. Cut and paste it to the Dashboard cell **E4**, then size it.
2. Untick **Legend** in Chart Elements (+).
3. Click the line > right-click > **Format Data Series** > **Fill & Line** > **Line**: colour `#1B6CA8`, width **2 pt**, tick **Smoothed line**; under **Marker** choose **None**.
4. Right-click the **vertical axis numbers** > **Format Axis** > **Axis Options** > Bounds **Minimum = 25** (makes the ups and downs easier to see).
5. Chart Elements (+) > **Axis Titles** > **Primary Vertical**, and type **No. of Patients**. Untick the horizontal axis title.



CHECKPOINT

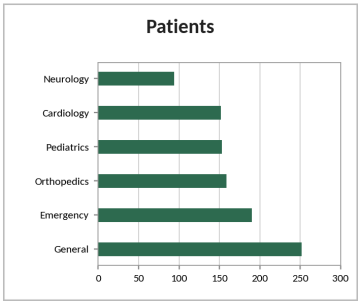
The line starts near **34** (Jan 2024), peaks at **57** in Nov 2025 and ends at 37. The axis labels show every second month.

STEP 19 Chart 2 – Patients by department (bar)

8 min

Select on Summary Data	Insert	Size (W × H)	Top-left at
A9:B15 (Department, Patients)	Charts > Insert Column or Bar Chart > **2-D Bar > Clustered Bar**	8.10 × 6.75 cm (3.19 × 2.66 in)	L4

1. Select **A9:B15**, insert a **Clustered Bar** (horizontal bars), cut and paste to **L4**, size it.
2. Untick **Legend**.
3. Colour the bars `#2D6A4F`.
4. Default gap width (150%) is fine.



GOOD TO KNOW

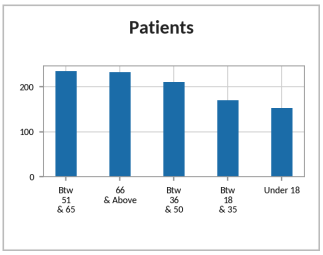
Bar charts draw the first row at the **bottom**, so **General (252)** is at the bottom and **Neurology (94)** on top. To flip: right-click the department names > Format Axis > tick **Categories in reverse order**. Your original keeps the default.

STEP 20 Chart 3 – Age groups (column)

8 min

Select on Summary Data	Insert	Size (W × H)	Top-left at
G9:H14 (Age Group, Patients)	Charts > Insert Column or Bar Chart > **2-D Column > Clustered Column**	7.08 × 5.48 cm (2.79 × 2.16 in)	E19

1. Select **G9:H14**, insert a **Clustered Column**, cut and paste to **E19**, size it.
2. Untick **Legend** and untick **Axis Titles** (a new chart shows placeholder "Axis Title" boxes; delete them).
3. Colour the columns `#1B6CA8` .



TIP

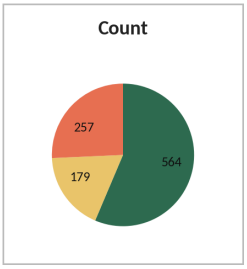
This chart, at 7.08 cm wide, slightly tucks under the pie chart. For a clean gap, make it **6.5 cm** (2.56 in) wide.

STEP 21 Chart 4 – Payment status (pie)

10 min

Select on Summary Data	Insert	Size (W × H)	Top-left at
J9:K12 (Payment Status, Count)	Charts > Insert Pie or Doughnut Chart > **2-D Pie**	5.07 × 5.48 cm (2.00 × 2.16 in)	H19

1. Select **J9:K12** (not column L), insert a **2-D Pie**, cut and paste to **H19**, size it.
2. Chart Elements (+) > **Data Labels**. Right-click a label > **Format Data Labels** > show **Value** (the counts 564, 179, 257).
3. Colour each slice: click the pie once, click the slice again to select just that slice, then Fill colour: **Paid** `#2D6A4F` , **Partially Paid** `#E9C46A` , **Pending** `#E76F51` .
4. Keep the **Legend** at the bottom (Chart Elements > Legend > Bottom) so readers know which colour is which; the finished workbook does this.



GOOD TO KNOW

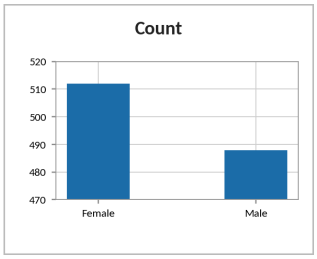
The heading says **(%)**, and because there are exactly 1,000 visits the counts are the percentages: 564 = 56.4%, 179 = 17.9%, 257 = 25.7%. To show real percentages, tick **Percentage** instead of Value in Format Data Labels.

STEP 22 Chart 5 – Patients by gender (column)

6 min

Select on Summary Data	Insert	Size (W × H)	Top-left at
N9:O11 (Gender, Count)	Charts > Insert Column or Bar Chart > **2-D Column > Clustered Column**	6.87 × 5.54 cm (2.70 × 2.18 in)	L19

1. Select **N9:O11**, insert a **Clustered Column**, cut and paste to **L19**, size it.
2. Untick **Legend** and **Axis Titles**.
3. Colour the columns `#1B6CA8` .



WATCH OUT

Excel may start this chart's vertical axis at **470** because the two values are close (512 and 488). That makes the gap *look* huge. Fix it honestly: right-click the axis > Format Axis > Bounds **Minimum = 0**. Bar charts should always start at zero.

CHECKPOINT

Female = **512**, Male = **488**. All five charts now sit inside their zones with no overlaps with the banner rows.

STEP 23 — Additional Insights banner and department table (B30:F37) — 15 min

1. Select **B30:O30** > **Merge & Center**. Type: **ADDITIONAL INSIGHTS ● Answers to remaining management questions**. Fill `#0B1C2C`, Arial 11 bold white.
2. In **B31:F31** type the five headings below. Fill `#1B6CA8`, Arial 8 bold white, centred.
3. In **B32:F37** link each cell to the Summary sheet. Type the formulas for **row 32**, then select B32:F32 and fill down to row 37:

B31	C31	D31	E31	F31
Department	Patients (Q2)	Total Bill (Q3)	Avg Stay (Q4)	Avg Ins % (Q7)
='Summary Data'!A10	='Summary Data'!B10	='Summary Data'!C10	='Summary Data'!D10	='Summary Data'!E10
(text)	number	$#,##0	0.0	0"%"

4. Format the body: Arial **8**, centred. **Zebra stripes:** fill rows **32, 34, 36** with `#F1F5F9` and leave rows 33, 35, 37 white. Number formats: **D** `$#,##0`, **E** `0.0`, **F** `0"%"`.
5. Add thin borders around B31:F37 in colour `#1E3A5F`.

WHY THIS WORKS

This is why column D was set to **11** in Step 13. At only 3 characters wide it would squeeze the Total Bill (Q3) numbers so they appear cut off (an earlier version of the dashboard had exactly that flaw).

CHECKPOINT

Row 32 shows: **General · 252 · $993,027 · 3.9 · 74%**. Row 37 shows **Neurology · 94 · $1,215,303 · 6.9 · 75%**. (Avg Ins 73.9 displays as 74%; 74.57 as 75%.)

STEP 24 — Q6 Financial Summary box (H31:J34) — 6 min

Cell	Content	Format
H31:J31 (merge)	Q6. Financial Summary	`#E76F51` fill · Arial 9 bold white, left-aligned
H32 / I32	Total Billed / **='Summary Data'!B3**	Labels: Arial 8. Values: Arial 8 **bold**, colour `#E76F51`, format `$#,##0`
H33 / I33	Paid / **='Summary Data'!B4**	same
H34 / I34	Outstanding / **='Summary Data'!B5**	same

1. Add thin borders `#1E3A5F` around **H32:I34**.

CHECKPOINT

Total Billed **$8,053,143** · Paid **$4,568,527** · Outstanding **$3,484,616**.

STEP 25 — Q10 Attention Areas box (L31:O36) — 8 min

1. Select **L31:O31** > Merge. Type **Q10. Attention Areas**. Fill `#2D6A4F`, Arial 9 bold white, left-aligned.
2. For **each** row 32, 33, 34, 35 and 36, select L:O of that row (for example L32:O32), click **Merge Cells**, and type one bullet line:

Row	Text to type (Arial 8, dark #0F172A, left-aligned, no fill)
L32	• $3.48M outstanding – prioritise collections
L33	• General & Emergency = highest volume
L34	• Orthopedics & Cardiology = highest revenue
L35	• Longer stays in Ortho (8.4d) & Neuro (6.9d)
L36	• 25.7% of bills still Pending

TIP

Typing the bullet and dash: Windows: hold `Alt` and type **0149** on the number pad for •, **0150** for –. Mac: `Option` + `8` for • and `Option` + `-` for –. Or use **Insert ▸ Symbol**.

These five lines are your **conclusions**. Each is backed by a number you built: $3.48M (Step 5), General & Emergency (Step 6), highest revenue and stays (Step 6), 25.7% Pending (Step 8).

STEP 26 Deeper Dive banner (B40:O40)

4 min

The second half of the dashboard is a row of four more charts. Rows 39 stays empty as a gap; the banner goes on **row 40**, the charts start on **row 41**.

1. Select **B40:O40** > **Merge & Center**. Type: **DEEPER DIVE • Billing, stay and insurance by department, plus billing trend**.
2. Format it exactly like the banner in row 30: fill `#0B1C2C`, Arial 11 bold white, centred. (Quickest: select B30, click **Format Painter**, then drag across B40:O40.)

GOOD TO KNOW

All four charts below are **6.5 cm wide × 5.5 cm high** (2.56 × 2.17 in), so they sit side by side in one row: top-left corners on **B41, E41, H41** and **L41**. They only use data that is already on your Summary sheet.

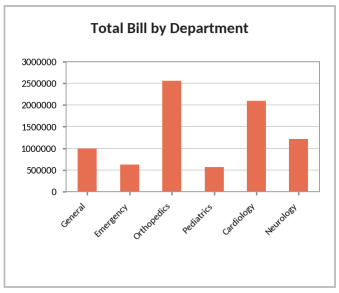
Selecting two separate blocks: for charts 6, 7 and 8 the department names are in column A but the numbers are in another column. Click **A9** and drag to **A15**, then hold `Ctrl` (Mac: `Cmd`) and drag over the number column **including its heading**. Both blocks stay highlighted.

STEP 27 Chart 6 – Total bill by department (column)

8 min

Select on Summary Data	Insert	Size (W × H)	Top-left at
A9:A15 + C9:C15 (Department, Total Bill)	Charts > Insert Column or Bar Chart > **2-D Column > Clustered Column**	6.5 × 5.5 cm (2.56 × 2.17 in)	B41

1. Select **A9:A15**, hold `Ctrl` and select **C9:C15**. Insert a **Clustered Column**, cut and paste it to **B41**, then size it.
2. Colour the columns `#E76F51`.
3. Click the chart title (it says *Total Bill*), type **Total Bill by Department** and press `Enter`.
4. Untick **Legend** if one appears. Axis Titles off.



CHECKPOINT

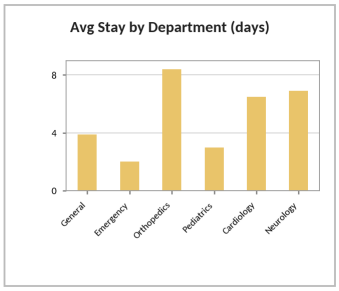
Orthopedics is the tallest column (about $2.55M), then Cardiology (about $2.10M); Pediatrics and Emergency are the shortest. This is the answer to management question 3.

STEP 28 — Chart 7 – Average stay by department (column)

6 min

Select on Summary Data	Insert	Size (W × H)	Top-left at
A9:A15 + D9:D15 (Department, Avg Stay)	Charts > Insert Column or Bar Chart > **2-D Column > Clustered Column**	6.5 × 5.5 cm (2.56 × 2.17 in)	E41

1. Select **A9:A15**, hold `Ctrl` and select **D9:D15**. Insert a **Clustered Column**, cut and paste it to **E41**, then size it.
2. Colour the columns `#E9C46A` .
3. Retype the title as **Avg Stay by Department (days)**.
4. Untick **Legend** and Axis Titles.



CHECKPOINT

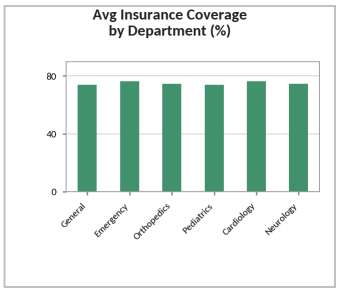
Orthopedics 8.4 days is the tallest, Neurology 6.9 next; Emergency is the shortest at 2.0 days. This answers question 4.

STEP 29 — Chart 8 – Insurance coverage by department (column)

8 min

Select on Summary Data	Insert	Size (W × H)	Top-left at
A9:A15 + E9:E15 (Department, Avg Insurance %)	Charts > Insert Column or Bar Chart > **2-D Column > Clustered Column**	6.5 × 5.5 cm (2.56 × 2.17 in)	H41

1. Select **A9:A15**, hold `Ctrl` and select **E9:E15**. Insert a **Clustered Column**, cut and paste it to **H41**, then size it.
2. Colour the columns `#40916C` .
3. Retype the title as **Avg Insurance Coverage by Department (%)**.
4. Right-click the vertical axis > **Format Axis** > Bounds **Minimum = 0** (and Maximum 90 if you like). Untick Legend and Axis Titles.



WHY THIS WORKS

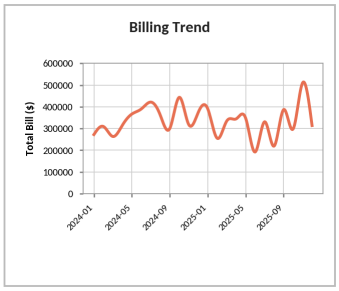
Coverage is 74–76% in every department, so the columns look almost equal. That *is* the finding: insurance coverage is consistent across departments. Starting the axis at 0 keeps the chart honest. This answers question 7.

STEP 30 — Chart 9 – Billing trend (line)

10 min

Select on Summary Data	Insert	Size (W × H)	Top-left at
A18:A42 + C18:C42 (Year_Month, Total Bill)	Charts > Insert Line or Area Chart > **2-D Line**	6.5 × 5.5 cm (2.56 × 2.17 in)	L41

1. Select **A18:A42**, hold `Ctrl` and select **C18:C42** (the Total Bill column). Insert a **2-D Line**, cut and paste it to **L41**, then size it.
2. Click the line > right-click > **Format Data Series** > **Fill & Line** > **Line**: colour `#E76F51` , width 2 pt, tick **Smoothed line**, Marker **None**.
3. Retype the title as **Billing Trend**.
4. Chart Elements (+) > **Axis Titles** > Primary Vertical: type **Total Bill ($)**. Untick Legend and the horizontal axis title.



CHECKPOINT

The line ends in a spike at **2025-11** (about $510,000), the busiest billing month. It rises and falls with the admissions chart, because more patients means more bills. This answers question 9.

STEP 31 Footer line (B53:O53) 3 min

1. Select **B53:O53** > Merge & Center. Type: **Data: Jan 2024 – Dec 2025 | 1,000 patient visits | Colour theme: Green ● Yellow ● Red ● Blue**
2. Fill `#0B1C2C`, Arial **8**, *italic*, white, centred.

GOOD TO KNOW

Why row 53? The Deeper Dive charts occupy rows 41 to about 51, so the footer closes the dashboard just beneath them.

STEP 32 Finish, check, save and share 15 min

Final quality check

Look at...	It should show
KPI cards	1,000 · $8,053,143 · 4.8 · $4,568,527 · 75%
Admissions chart	Line from about 34 up to a peak of 57 in 2025-11
Department bars	General 252, Emergency 190, Orthopedics 159, Pediatrics 153, Cardiology 152, Neurology 94
Age-group columns	235 · 232 · 210 · 170 · 153
Payment pie	564 Paid (dark green), 179 Partially Paid (yellow), 257 Pending (coral)
Gender columns	Female 512, Male 488
Q6 box	$8,053,143 · $4,568,527 · $3,484,616
Q10 box	Five bullets, readable, none cut off
Deeper Dive row	Four charts in a row: Total Bill (orange), Avg Stay (yellow), Avg Insurance (green), Billing Trend (orange line)
Chart count	**9 charts** in total (5 above the insights banner, 4 in the Deeper Dive row)

1. Click cell **A1** so the sheet opens clean, set the zoom to a comfortable level (80–100%), and make sure the **Dashboard** tab is the one showing when you save.
2. **Print / PDF set-up:** Page Layout ▸ Orientation ▸ Landscape, then in **Scale to Fit** set **Width = 1 page** and **Height = 1 page**. Export with File ▸ Export ▸ Create PDF/XPS.
3. **Save** (`Ctrl` + `S`).
4. **Make the picture for your GitHub README:** bring the whole Dashboard into view and press `Windows` + `Shift` + `S` (Mac: `Cmd` + `Shift` + `4`), drag around the dashboard, and save it as **Screenshot_of_Dashboard.png**. Replace the old file in your repository.

CHECKPOINT

Congratulations – you have rebuilt the Secure Life General Hospital Performance Dashboard with all **9 charts**, exactly as described in your project write-up, and every number traces back to your data.

PART 5

Practice and Level Up

Your practice plan, what to learn next, and how to publish your work.

Optional · at your own pace

Your practice kit and practice plan

Your project folder contains a **Practice_Kit** so you can rebuild the dashboard as often as you like:

File	What it is	How to use it
Practice_Workbook_START_HERE.xlsx	A workbook with the three sheets already created and the 1,000 raw records in **Patient Data** (columns A–K). Steps 1 and 2 are done.	Start at **Step 2** (styling) or **Step 3**, then continue to Step 32. Never practise inside the finished workbook.
hospital_patient_data_raw.csv	The same 1,000 records as plain text.	Practise importing data: **Data** ▸ **From Text/CSV** (Step 2).
SECURE_LIFE_GENERAL_HOSPITAL_Dashboard.xlsx	The finished, corrected workbook.	Compare your results with it, cell by cell.

Round	What to do	Goal
1 Follow	Work through Steps 3 to 32 with this guide open.	A finished dashboard that passes every CHECKPOINT.
2 Repeat	A day later, restart from the Practice Workbook. Use only the cell map (Part 1) and Appendix B.	Finish without reading the step text.
3 Explain	Without notes, explain aloud what COUNTIF, SUMIF, AVERAGEIF and SUMPRODUCT do.	Be ready for interview or presentation questions.
4 Extend	Add one new insight of your own (for example average bill per department) and a chart for it.	Show initiative beyond the brief.

Where to go next

Skill	What it gives you	How to start
Slicers	Real clickable filters (pick a department or year and every chart changes).	Build the Summary tables as **PivotTables**, then **PivotTable Analyze** ▸ **Insert Slicer**.
Excel Tables	Formulas that grow automatically when you add new rows.	Select the data > `Ctrl` + `T` ; then use names like *Table1[Total Bill Amount]* in formulas.
Conditional formatting	Colour cells automatically (for example highlight high bills).	**Home** ▸ **Conditional Formatting**.
Power Query	Clean and combine data from CSV files with no copy-pasting.	**Data** ▸ **Get Data**.
Publishing	Share your work with the world.	Upload the workbook, this guide and your screenshot to GitHub (see below).

Updating your GitHub repository

1. Open your repository in a browser (not the phone app): **github.com/eunicemarf89/Secure-Life-General-Hospital-**.

2. Tap **Add file > Upload files**, add your **new screenshot** (same name replaces the old one) and this guide, then **Commit changes**.

Appendix A · Colours and fonts

Every colour used, with its Hex code and RGB values (use RGB if your Excel has no Hex box). Font everywhere on the Dashboard: **Arial**.

Swatch	Hex	RGB	Name	Used for
	#0B1C2C	11, 28, 44	Navy	Title banner, section headers, footer, Diagnosis header
	#0D1B2A	13, 27, 42	Deep navy	Patient Data header row; Department/Year header cells
	#1B6CA8	27, 108, 168	Blue	KPI 1, strip (row 2), Year chips, table headers, line & column charts
	#2D6A4F	45, 106, 79	Dark green	KPI 2, Emergency chip, department bars, Paid slice, Q10 header
	#E9C46A	233, 196, 106	Yellow	KPI 3 card, Partially Paid slice
	#E76F51	231, 111, 81	Coral	KPI 4 card, Pending slice, Q6 header, Q6 values
	#40916C	64, 145, 108	Green	KPI 5 card
	#F4A261	244, 162, 97	Sand orange	“General” chip
	#E0F2FE	224, 242, 254	Sky	Diagnosis list cells
	#F1F5F9	241, 245, 249	Light grey	Zebra rows in the department table
	#0F172A	15, 23, 42	Ink	Dark text on yellow / sky / white
	#1E3A5F	30, 58, 95	Border blue	Thin borders around tables and chips

Appendix B · Formula cheat sheet

Where	Formula
Patient Data L2	`=F2-E2`
Patient Data M2	`=YEAR(E2)`
Patient Data N2	`=TEXT(E2,"yyyy-mm")`
Patient Data O2	`=IF(C2<18,"Under 18",IF(C2<=35,"Btw 18 & 35",IF(C2<=50,"Btw 36 & 50", IF(C2<=65,"Btw 51 & 65","66 & Above"))))`
Summary B2	`=COUNTA('Patient Data'!$A$2:$A$1001)`
Summary B3	`=SUM('Patient Data'!$I$2:$I$1001)`
Summary B4	`=SUMIFS('Patient Data'!$I$2:$I$1001,'Patient Data'!$K$2:$K$1001,"Paid")`
Summary B5	`=B3-B4`
Summary B6	`=ROUND(AVERAGE('Patient Data'!$L$2:$L$1001),1)`
Summary B7	`=ROUND(AVERAGE('Patient Data'!$J$2:$J$1001),0)`
Summary B10	`=COUNTIF('Patient Data'!$G$2:$G$1001,$A10)`
Summary C10	`=SUMIF('Patient Data'!$G$2:$G$1001,$A10,'Patient Data'!$I$2:$I$1001)`
Summary D10	`=ROUND(AVERAGEIF('Patient Data'!$G$2:$G$1001,$A10,'Patient Data'!$L$2:$L$1001),1)`
Summary E10	`=ROUND(AVERAGEIF('Patient Data'!$G$2:$G$1001,$A10,'Patient Data'!$J$2:$J$1001),1)`
Summary H10	`=COUNTIF('Patient Data'!$O$2:$O$1001,$G10)`
Summary K10	`=COUNTIF('Patient Data'!$K$2:$K$1001,$J10)`
Summary L10	`=SUMIF('Patient Data'!$K$2:$K$1001,$J10,'Patient Data'!$I$2:$I$1001)`

Where	Formula
Summary O10	`=COUNTIF('Patient Data'!$D$2:$D$1001,$N10)`
Summary A19	`=TEXT(EDATE(DATE(2024,1,1),ROW()-19),"yyyy-mm")`
Summary B19	`=SUMPRODUCT(--('Patient Data'!$N$2:$N$1001=$A19))`
Summary C19	`=SUMPRODUCT(('Patient Data'!$N$2:$N$1001=$A19)*'Patient Data'!$I$2:$I$1001)`
Dashboard cards	`='Summary Data'!B2 (B3 · B6 · B4 · B7 for cards 2–5)`
Dashboard B32:F32	`='Summary Data'!A10 … E10, filled down to row 37`
Dashboard I32:I34	`='Summary Data'!B3 · B4 · B5`

Appendix C · Answer key

Use this page to verify your own workbook. All values are taken from the finished file.

Headline	Value
Total patient visits	1,000
Total billed	$8,053,143.44
Paid	$4,568,527.32
Outstanding	$3,484,616.12
Average stay	4.8 days
Average insurance	75%
Payment mix	Paid 564 (56.4%) · Partially Paid 179 (17.9%) · Pending 257 (25.7%)
Gender	Female 512 · Male 488
Age groups	Btw 51 & 65: 235 · 66 & Above: 232 · Btw 36 & 50: 210 · Btw 18 & 35: 170 · Under 18: 153
Years	2024: 508 visits · 2025: 492 visits

Department answer key

Department	Patients	Total Bill	Avg Stay (days)	Avg Insurance %	Dashboard shows
General	252	993,026.92	3.9	73.9	$993,027 · 74%
Emergency	190	622,525.28	2.0	76.2	$622,525 · 76%
Orthopedics	159	2,553,862.32	8.4	74.5	$2,553,862 · 74%
Pediatrics	153	572,987.96	3.0	74.0	$572,988 · 74%
Cardiology	152	2,095,437.97	6.5	76.3	$2,095,438 · 76%
Neurology	94	1,215,302.99	6.9	74.6	$1,215,303 · 75%

Key findings these numbers support

Finding	Evidence
Busiest departments	General (252) and Emergency (190)
Highest revenue	Orthopedics ($2.55M) and Cardiology ($2.10M) despite fewer patients
Longest stays	Orthopedics 8.4 days, Neurology 6.9 days; shortest: Emergency 2.0
Cash-flow exposure	$3.48M outstanding of $8.05M billed
Older patients	51+ account for 235 + 232 = 467 visits, over 45% of all visits
Peak month	November 2025: 57 admissions, $510,414 billed

Appendix D · When something goes wrong

What you see	Why it happens	How to fix it
##### in a cell	The column is too narrow for the number.	Widen it (double-click the right edge of the column letter) or lower the font size.
Stay_Length shows a date like 1900-01-08	Excel copied the date format from the cells it subtracted.	Home > Number Format box > **General**.
#NAME?	A function name is misspelled, or text is missing straight quotes.	Retype the function name; wrap text in `"` quotes.
#REF!	A cell or sheet the formula pointed to was deleted.	Press Undo; check the sheet is still named exactly *Patient Data*.
#VALUE!	A formula tried to do maths on text (often text dates).	Check dates are right-aligned (real dates); see next row.
Dates are left-aligned	They were imported as text.	Select the column > Data > **Text to Columns** > **Finish**; or re-copy from the source.
COUNTIF/SUMIF gives **0**	The label you look up is spelled differently from the data (extra space, capital letter).	Retype the label, or use the Data > Filter drop-down to see the exact spelling.
Months turned into dates (Jan-24)	Excel "helpfully" converted 2024-01.	Format the cells as **Text**, or type an apostrophe first: `'2024-01`
Formula appears as text, not result	Cell is formatted as Text, or the formula started with a space.	Set the format to General, click into the cell and press Enter.
Chart shows two lines/columns	Excel treated the label column as data.	Right-click the chart > **Select Data** and keep only the numeric series.
Chart did not change after edits	Calculation is set to manual.	Formulas > Calculation Options > **Automatic**.
Card shows 0.75 instead of 75%	Number format not set.	Ctrl+1 > Custom: `0"%"`
Comma formula rejected	Your region uses semicolons.	Use semicolons: `=IF(A1>1;"a";"b")`

Appendix E · Shortcuts and glossary

Windows	Mac	Does
`Ctrl + S`	`Cmd + S`	Save
`Ctrl + Z` / `Ctrl + Y`	`Cmd + Z` / `Cmd + Y`	Undo / Redo
`Ctrl + C` / `X` / `V`	`Cmd + C` / `X` / `V`	Copy / Cut / Paste
`Ctrl + 1`	`Cmd + 1`	Format Cells (number formats, borders, shrink to fit)
`Ctrl + T`	`Ctrl + T`	Turn a range into a Table
`Ctrl + B`	`Cmd + B`	Bold
`F4`	`Cmd + T`	Cycle $ (absolute) references while editing a formula
`F2`	`Ctrl + U`	Edit the selected cell
`Ctrl + Arrow`	`Cmd + Arrow`	Jump to the edge of the data
`Ctrl + Shift + Arrow`	`Cmd + Shift + Arrow`	Select to the edge of the data
`Ctrl + Home`	`Cmd + Fn + ←`	Go to A1
`Alt + Enter`	`Ctrl + Option + Enter`	New line inside a cell

Glossary

Absolute reference – A cell reference with $ signs that does not move when copied.

Helper column – An extra column you add to make later calculations easy.

Aggregate – A single number summarising many rows (sum, count, average).	**KPI** – Key performance indicator – a headline number.
Category axis – The axis of a chart that shows names (departments, months).	**Merged cells** – Several cells joined into one.
Dashboard – One screen that summarises the key numbers and charts of a project.	**PivotTable** – A tool that counts, sums and groups data without formulas.
Data label – The number printed on a bar, slice or point.	**Range** – A block of cells such as A2:A1001.
Fill handle – The small square at a cell's corner used to copy it down.	**Series** – One set of values plotted in a chart.
Formula – A calculation that starts with = .	**Sheet** – One tab inside a workbook.
Function – A built-in formula such as SUM or COUNTIF.	**Table** – A range Excel treats as a structured list with a styled header.

GOOD TO KNOW

Final reminder: the dashboard is only as good as the checks behind it. Run the CHECKPOINT boxes, compare against Appendix C, and your numbers will always be defensible when someone asks "where does this figure come from?"